

zamiaCAD: An Open Source Platform for Advanced Hardware Design

Computer-aided design (CAD) software basically makes use of a computer to assist in the creation, optimisation, analysis or modification of a design. From increasing the productivity level of the designer to improving the quality of designs to creating and maintaining a database for manufacturing, CAD software are used extensively across many verticals. Let's discuss more about zamiaCAD, which is included in this month's EFY DVD

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ZamiaCAD is a project which aims at transferring the advances in software development to the hardware development discipline. The zamiaCAD project targets to support state-of-the-art and future large-scale industrial hardware designs. This framework is developed to provide the research community with a robust open source research platform that is equipped with powerful debugging functionality.

Written in Java, zamiaCAD is an extensible and modular platform for research, analysis and advanced hardware design. Its core components are language-dependent frontends generating an instantiation graph (IG), applications working on the IG data structure and the language-independent IG data structure. The frontends consist of an elaboration engine and a parser. Currently, Verilog has only a parser, whereas VHDL has a complete frontend. There are also a number of video tutorials available on the website to give the users an idea of how zamiaCAD looks and feels like.

Basic build process

zamiaCAD has a complete VHDL compiler with project build facilities. The entire build process is actuated by

the BuildPath.txt file, which informs the builder about the location where he/she can find the sources and libraries to put them into. Since sources may be stored on high-latency network mount points (for example NFS or AFS mounts), zamiaCAD has a built-in file system cache (FSCache), which it uses to cache files and directory information on the user's local disk for speed.

Parsing all VHDL sources that are accessible through the directories configured in a project's BuildPath.txt can be very time consuming. Since a significant portion of the design units defined in those files might never actually get instantiated or used in the project (this is especially true for technology/simulation libraries), it is often not necessary. Therefore zamiaCAD comes

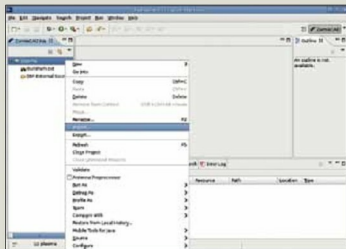


Fig. 1: Import dialogue



Fig. 2: Navigating through zamiaCAD

with a very high-speed VHDL indexer, which will extract only the information about which design units are declared